



Armadale Kalamunda Group

Full Neurological Observations Guideline

1. Purpose

This guideline outlines how to perform full neurological observations.

Initiate Escalation of Care Process and activation of Medical Emergency Team (MET) if patient meets criteria.

2. Applicability

This guideline applies to all medical, nursing, midwifery and allied health staff at Armadale Kalamunda Group.

3. Guidelines

3.1 Alert / vocal (responds to) / painful stimuli (responds to) / unresponsive (AVPU)

The AVPU method provides an initial rapid neurological assessment to determine conscious level. The AVPU assessment is to be performed as part of a patient's physiological observations.

Consider Escalation of Care Process and Medical Emergency Team (MET)

3.2 Glasgow Coma Scale (GCS)

If AVPU is abnormal, perform and document GCS. The score attained from the GCS provides an initial assessment and baseline comparison with future scores to determine improving, deteriorating or static clinical states.

The GCS evaluates two areas of consciousness, namely arousal and cognition and the score derived from the patient's response in three key.

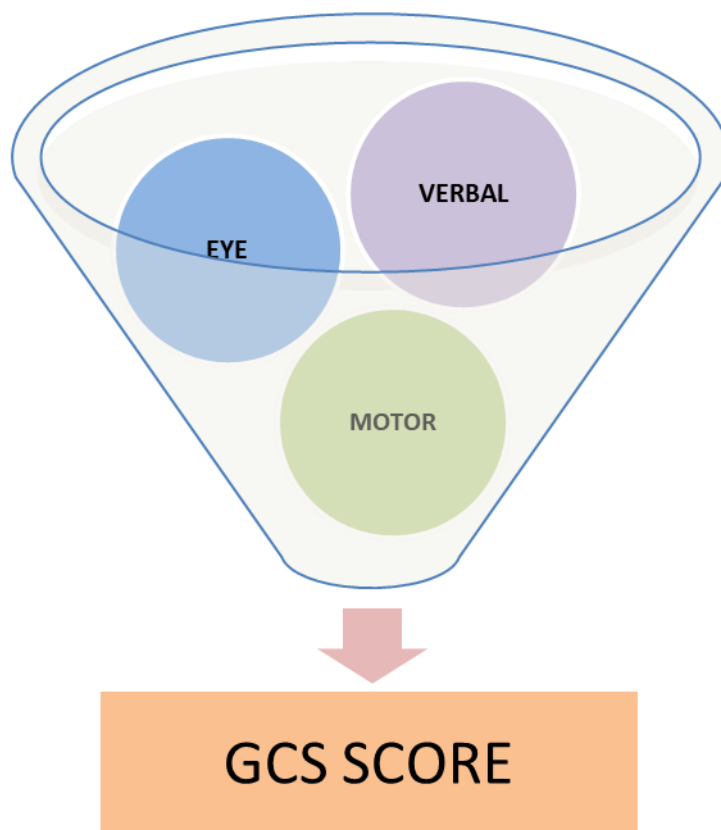
3.3 Pre assessment considerations

- Ascertain if English is the primary language and engage an interpreter if required.
- The GCS score may be affected if the patient is under the influence of substances e.g. alcohol or medication.
- Communication barriers may be present e.g. aphasia, eyelid dyspraxia, hearing or visual impairment and may require alteration to the assessment techniques used. Acute aphasia disorders usually develop quickly as a result of head injury or stroke, whereas progressive aphasia develops slowly from a potential brain tumour, infection or dementia. The symptoms

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and type of aphasia is determined by the extent of brain damage or atrophy sustained. It is important that the healthcare practitioner documents any communication barriers that may impact the assessment or GCS score.

- Medical Devices e.g. Tracheostomy, Endotracheal Tube (ETT) must be denoted with a “T” on the relevant chart
- Environment should have limited background noise and distractions.
- Consistency between users is imperative as irregular scoring may potentially lead to a compromise in patient safety.
- If appropriate, obtain patient history from relatives, Paramedics, Police, General Practitioner, carers or friends, to ascertain any changes in patients:
 - Mood
 - Intellect
 - Memory and Personality
 - Behavior



3.4 Contraindications for procedure

Chronic illness, disability, acute trauma and oedema can lead to inaccurate neurological assessment and thus an inaccurate GCS score.

Constraints to the use of the GCS tools should be considered and barriers clearly documented. If one of the three category components are not measurable the GCS cannot be used.

3.5 Physiological observations

Observe and assess if the patient's behaviour appears appropriate on initial interaction.

The following physiological observations will be taken and recorded at a frequency determined by the MO. Document results on both the Neurological Examination Chart and the Adult Observation and Response Chart (AORC).

- | | |
|---|--|
| <ul style="list-style-type: none"> • Blood Pressure • Heart Rate • Respiration | <ul style="list-style-type: none"> • Temperature • Oxygen Saturation Levels • Blood Glucose Level |
|---|--|

Record cardiac rhythm if monitored and consider 12 lead ECG.

Blood pressure and heart rate changes can indicate raised intra-cranial pressure leading to irreversible brain stem damage. Cushing's Triad is a late sign and emergency action should be initiated. Initiate Escalation of Care Process and Activation of Medical Emergency Team (MET).

3.6 Eye response

Assess the patient's ability and ease of rousability. Observe for eye swelling and if present mark the site specific form corresponding to 'none' with a 'C' to denote the eye is 'closed'.

Open Spontaneously (Score 4)	This observation is made without touching the patient
Opening to Verbal Command (Score 3)	Instruct the patient to open their eyes in a firm loud voice
Opening to Pain (Score 2)	Apply a (peripheral) painful stimulus to elicit the eye-opening response <i>only</i> if the patient does not respond to any other stimuli (see 3.13 Peripheral painful stimuli)
None (Score 1)	If the patient's eyes have been open continuously (no blinking observed) record as 'no eye-opening response' Observe that the patient does not open their eyes to any stimuli

3.7 Verbal response

Assess the patient's ability to respond appropriately to questions posed to them. Only ask one question at a time and always allow adequate time for the patient to respond to your questions. Speak in a voice loud enough for the patient to hear, varying the questions. If the patient is unable to vocalise due to the presence of a tracheostomy or endotracheal tube, document this with a "T" in the "None" column. If the patient is aphasic document this factor with the letter "A".

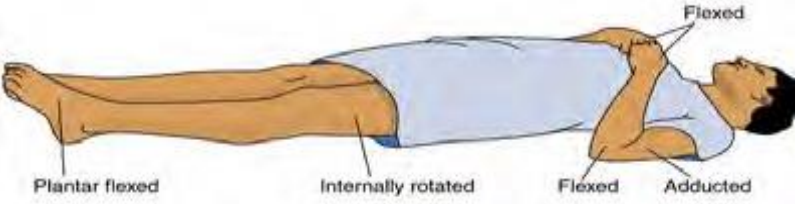
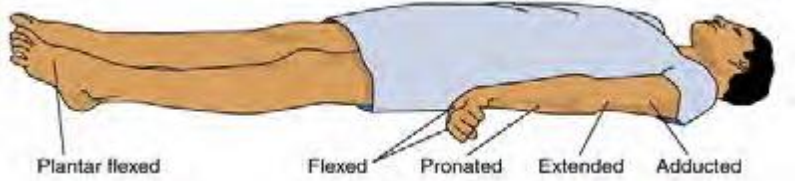
Orientated Response (Score 5)	Ask the patient one question related to each of the following Time
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	Place Person
Confused Response (Score 4)	The patient is able to talk but is not orientated
Inappropriate Response (Score 3)	Observe the patients disorganized or inappropriate speech Consider patient history
Incomprehensible Response (Score 2)	Observe the patient making unintelligible sounds
None (Score 1)	Observe the patient making no sounds or speech even with stimulation

3.8 Best motor response

This measures the overall awareness and ability to respond by movement to external stimuli. Motor responses can be affected by medications e.g. muscle relaxants and or sedation. This should be recorded on the site specific forms for patients with impaired consciousness as follows - M = Muscle Relaxant S = Sedation

Obeys commands (Score 6)	Assess patient response to simple commands for movement and allow a short time for response. If the patient does not respond to any verbal instructions, apply painful stimuli to elicit a response.
Localises to pain with purposeful movement (Score 5)	Observe the patient attempting to remove or locate the painful stimuli in a purposeful way.
Withdraws from pain (Score 4)	Observe the patient attempting to withdraw the limb away from the painful stimuli but does not attempt to remove the stimuli or locate the source of the pain.
Flexion to pain (Score 3) Abnormal flexion and decorticate (flexor)	Observe the patient performing non-purposeful flexion to painful stimuli. This presents as flexion of the elbows, wrists and fingers with the arms held tightly to the sides and extension, internal rotation of the legs. The feet are plantar flexed.

	 Decorticate Posturing This response correlates with damage to the cerebral hemispheres interrupting the cortico - spinal pathways of diencephalon.
Extension to pain (Score 2) Abnormal extension and Decerebrate (extensor)	Observe if the patient presents with extension of the neck adduction of the arms and extension of the elbows, pronation of the forearms with the wrists and fingers flexed to painful stimuli. The legs are also extended and feet plantar flexed (with the toes pointing downwards) ¹. This response may indicate damage to the midbrain and upper pons. It is a sign of greater dysfunction than decortication. A change from decortication (flexor) to decerebration (extensor) may indicate extension of the lesion into the upper brain stem.  Decerebrate Posturing
None (Score 1)	Observe no response to painful stimuli.

3.9 Findings

The GCS should ideally be reported in its component parts with an overall score so as to identify the area of deficit if present.

Initiate Escalation of Care Process and Activation of Medical Emergency Team (MET) if there is a sudden drop in the GCS by two points or more.

A GCS of 3 is the lowest possible score and indicates total unresponsiveness, whilst a maximum score of 15 indicates that the client is alert, awake and responsive. A GCS of 8 or less may require respiratory support.

Best eye response	
1	None
2	Open to Pain
3	Open to Verbal Command
4	Open Spontaneously

Best verbal response	
1	None
2	Incomprehensible Response
3	Inappropriate Response
4	Confused Response
5	Orientated Response

Best motor response	
1	None
2	Extension to Pain
3	Flexion to Pain
4	Withdraws from Pain
5	Responds to Pain with Purposeful Movement
6	Obeys Commands

If GCS <8 initiate Escalation of Care Process and consider MET

3.10 Assessment of limb symmetry, tone and strength

The limbs and trunk should be assessed for any indication of asymmetry, poor tone or muscle wasting.

Asymmetry	Each limb has to be individually assessed as appropriate. Record right and left limb movement separately and any differentiation.
Tone	This is assessed through putting the limbs through a range of passive movement.
Strength	Requesting the patient to push against your resistance.

Document the best response for each limb separately. Use “R” to denote the right side and “L” to denote the left side. Equal movement is recorded with a “●”.

3.11 Assessment of pupil size and reaction to light



- Perform assessment on both eyes, although conditions such as swelling, and trauma may not permit this.
- Pupil size is measured by comparing it with a series of circular millimetre measures on the chart ¹. The size of the pupil is controlled by the third cranial nerve (oculomotor nerve). This nerve runs over the surface of the brain and in the presence of localized swelling may be compressed. The result is a decrease in nerve function, relaxation of muscle and dilation of the pupil
- Progressive dilation and loss of pupil reaction on one side may indicate an enlarging intracranial mass and will require further investigation.

- Attempt to darken the room where possible as this allows for the best response to occur. Pupils are normally black, round and equal in diameter (average 2mm-5mm).
- If the patient is unable to open their eyes, consider manually opening the eyelid for them
- Ask the patient to stare at a distant object.
- Observe the patient's pupil size, shape and position prior to light stimulation
- Shine the pen torch from the outer aspect of the eye to the pupil
- Assess for:
 - Speed of papillary reaction and size of pupil when constricted
 - Consensual (both pupils) reaction to the light stimulus
 - Pupil reaction is documented as
 - + = reacts
 - - = no reaction
 - C = eye closed

3.12 Painful stimuli

- All painful stimuli applied must stop following a response from the patient and will not continue for longer than 10 seconds.
- Peripheral stimuli must be used first.
- If there is no response to using peripheral stimulus, then central stimulus may be used.
- Patient response and type of painful stimuli applied to be documented.
- Painful stimuli will only be used when absolutely necessary and should not become routine.

3.13 Peripheral painful stimuli

Direct pressure to the nailbed is contraindicated as it will damage the capillary bed with possible loss of the nail.

- Use only a pen or pen light to apply pressure to the side (lateral outer aspect) of the nail bed. Repeat until all limbs have been tested as they may all respond differently.

3.14 Central stimuli

The sternal rub and supraorbital pressure are not to be used as they may cause harm to the patient. Localised soft tissue damage and displacement fractures have been reported.

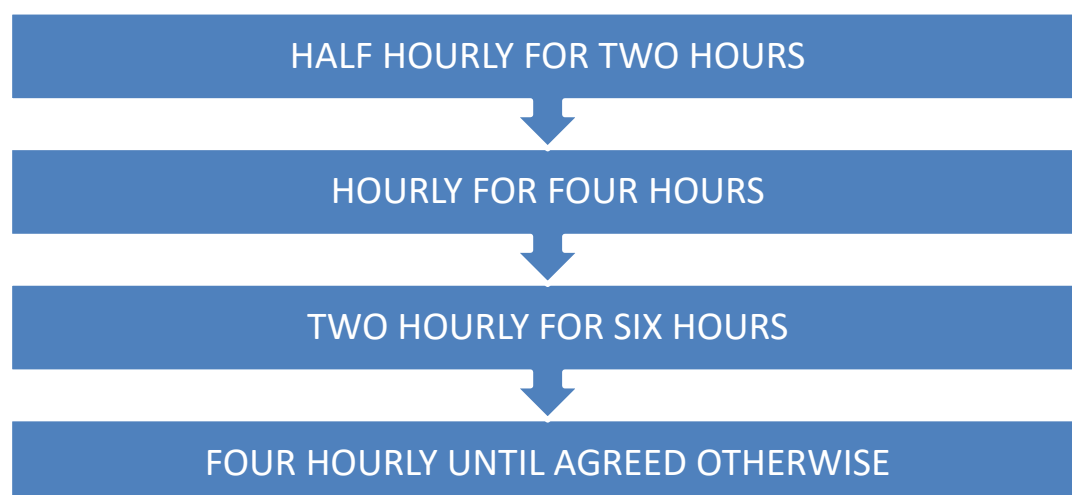
Central stimuli can be performed as appropriate using the following:

Trapezium Squeeze	• Confirm the patient has no cervical injuries • Locate the trapezius muscle on the patient and squeeze firmly as if giving a hard massage
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3.15 Frequency of observation following head injury

The risk of rapid patient deterioration is highest during the first 6 hours of sustaining a head injury. An individual who requires hospital admission after the initial assessment in the Emergency Department, should receive the following observations at a minimum and frequency increased on an individual basis as requested by the MO.

Suggested frequency of Full Neurological Observations



It is a requirement that medical staff review the patient, document a plan of care prior to admission and review progress at least once within 24 hours.

4. Legislative context

The following documents are required to give effect to this policy:

- Health Department of Western Australia (HDWA) [Recognising and Responding to Acute Deterioration Policy MP 0171/22](#)
- HDWA [Clinical Handover Policy MP 0095](#)
- East Metropolitan Health Service (EMHS) [Patient Identification Policy EMHS: 21](#)

5. Supporting information

The following documents support the implementation of this policy:

- [Pain Management Procedure AKG-PRO-0339-17](#)
- [Neonatal: Routine and Additional Observations Guideline PAE-GUI-0288-16](#)
- [Interpreter and Translator Guidelines AKG-GUI-0212-00](#)
- [Recognition and Response to Acute Deterioration Policy AKG-POL-0388-14](#)

6. Compliance, monitoring and evaluation

Compliance with this policy will be monitored via:

- Monitoring of reported clinical incident data via Datix CIMS.

Department specific findings are to be reported at the relevant departmental meeting for review and identification of any opportunities for improvement.

7. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 6 – Action 6.5 Correct identification and procedure matching
- Standard 6 – Action 6.7 Clinical handover
- Standard 8 – Action 8.3 Partnering with consumers
- Standard 8 – Action 8.4 Recognising acute deterioration
- Standard 8 – Action 8.6 Escalating care
- Standard 8 – Action 8.10 Responding to deterioration

8. Document owner

Director Nursing and Midwifery

Enquiries relating to this guideline may be directed to:

Title: Nurse Coordinator
 Department: Hospital Wide
 Email: jane.husain@health.wa.gov.au

9. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V4	06/05/2024	06/05/2027	

10. Authorisation

This guideline has been authorised and issued by the Director Nursing and Midwifery.

Authorised by	Helen Fullarton – Director of Nursing and Midwifery
Authorisation date	06/05/2024
Published date	09/05/2024

Appendix 1: General information

There are **two** primary functions in neurological assessment:

1. Determine current neurological status
2. Assessment of changed or impaired level of consciousness and coma due to a wide range of conditions

Full Neurological Observation incorporates the following patient assessments:

- **Alert, Vocal, Painful Stimuli, Unresponsive to Stimuli - AVPU** (Quick Check)
- Glasgow Coma Scale (Ongoing Assessment)
- Monitoring of physiological observations
- Observation of the following focal neurological signs including:
 - Pupil size and reaction to light
 - Motor function
 - Sensory Function
 - Mental Status

Through accurate initial and ongoing assessment of the patient, staff are in a position to initiate an escalation pathway if required to improve the outcome and reduce long term consequences for the patient.

Standardised neurological assessment tools provide the ability for findings to be more accurately reported and therefore improve communication between healthcare professionals and provide optimum provision of clinical care for the patient, particularly in the event of a critical situation.

Causes of Altered Neurological Status

Altered neurological status can be consequential to multiple causes. The diagram below provides examples but is by no means exhaustive:

Structural Insults Haematoma Cerebral Aneurysm Stroke Cerebral Tumour	Metabolic Insults Loss of substrate Shock – Hypovolaemic Hypoxia Hypoglycaemia Cardiogenic Focal Ischaemia – Stroke
Abnormal Pathology Hypo or Hyponatraemia Hyperosmolarity Hyperglycaemia Seizures Mental illness Hypercalcaemia	External Trauma Alcohol consumption Hypothermia Heat Exhaustion Drugs – illicit/prescription Envenomation

When to Perform Full Neurological Observations

- History of head trauma
- Toxicology (alcohol or drug ingestion)

- Episode of collapse, unwitnessed fall, or loss of consciousness
- Evidence of a change in level of consciousness
- Neurological intervention/surgery
- When a patient requires sedation

Neurological Symptoms

These may include but not be exhaustive of the following.

Headache	Fevers	Changes to memory
Nausea or vomiting	Alterations to bladder or bowel function	Syncope/Change in level of consciousness
Seizures	Changes in vision, hearing, speech	Alteration in swallow function
Numbness and paresthesia	Changes in Mood and Sleep Pattern	Paralysis/Weakness